EMMA: Emerging Medicine Mentoring Agent

by

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- The developers of the BERTopic [4] and Surprise [5] libraries for enabling advanced clustering and recommender system implementations

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List of Abbreviations

AI Artificial Intelligence 11, 12

AMG-RAG Agentic Medical Graph-RAG 2, 3, 6, 19, 20, 26

BOW Bag of Words 11, 12

CF Collaborative Filtering 1, 4, 17, 18, 25

EMMA Emergency Medicine Mentoring Agent 1–3, 10, 12, 14, 17, 21–23

FAISS Facebook AI Similarity Search 1, 3, 6, 8, 9, 19, 23

GMM Gaussian Mixture Model 15

GPU Graphics Processing Unit 4

HDBSCAN Hierarchical Density-Based Spatial Clustering of Applications with Noise 14, 15, 22

HR@K Hit Rate@K 18

IndexFlatIP Index Flat Inner Product 3, 19

KNNBasic K-Nearest Neighbors Basic 17, 18, 26

LinearSVC Linear Support Vector Classification 1, 4, 12, 21, 24, 26

LLM Large Language Model 1, 3, 4, 6, 8, 19, 20, 23, 26

LOO Leave-One-Out 17

LSA Latent Semantic Analysis 15

ML Machine Learning 1, 2

MNB Multinomial Naïve Bayes 12

NER Named Entity Recognition 1, 2, 8, 9, 23, 25, 26

NF4 4-bit NormalFloat 4

NMF Non-negative Matrix Factorization 17

P@K Precision@K 18, 25

PMI Pointwise Mutual Information 3, 9

pp percentage points 1, 4, 6, 19, 20, 23, 25, 26

RAG Retrieval-Augmented Generation 1, 3, 4, 6, 9, 10, 19, 20, 22–26

RMSE Root Mean Square Error 18

SVD Singular Value Decomposition 17, 18

TF-IDF Term Frequency-Inverse Document Frequency 1, 4, 11, 12, 15, 21, 22, 24, 26

UMAP Uniform Manifold Approximation and Projection 14, 16

US-MLE United States Medical Licensing Examination 1, 2, 6, 8, 9

VRAM Video Random Access Memory 4, 6, 26

Chapter 1

Introduction

Medical licensing examinations such as the United States Medical Licensing Examination (US-MLE) represent one of the most demanding knowledge challenges in professional education. Candidates must synthesize clinical reasoning across dozens of specialties under timed, high-stakes conditions. Traditional preparation relies on self-directed study, peer quizzing, and expensive tutoring. These resources are not equally accessible to all candidates.

This project presents Emergency Medicine Mentoring Agent (EMMA), a conversational medical study agent built to close that gap. In explanation mode, students pose clinical questions in natural language and receive responses grounded in passages from eighteen standard medical textbooks. In quiz mode, EMMA presents authentic US-MLE-style questions, evaluates answers, and tracks per-specialty performance over time. A collaborative filtering recommender then steers students toward their weakest areas. EMMA is built on a five-component Machine Learning (ML) pipeline: a SpaCy Named Entity Recognition (NER) model for clinical entity extraction, a Facebook AI Similarity Search (FAISS) vectorstore over 36,723 textbook chunks, a Term Frequency-Inverse Document Frequency (TF-IDF) + Linear Support Vector Classification (LinearSVC) specialty classifier, a BERTopic clustering model for fine-grained topic routing, and a Surprise-based Collaborative Filtering (CF) recommender. All components feed a locally-run Qwen3-4B-Thinking-2507 Large Language Model (LLM). The central empirical question is whether textbook-grounded Retrieval-Augmented Generation (RAG) improves a small local LLM's accuracy on MedQA compared to a no-retrieval baseline. A six-run ablation grid shows the answer is condition-dependent: with the right embedding and LLM pairing, RAG delivers an +11 percentage points (pp) accuracy gain; without those conditions, it can hurt. This nuanced finding is the primary contribution of this report.

Chapter 2

Problem Formulation

Medical exam preparation is a laborious task requiring extensive rote practice to internalise clinical vocabulary, symptom presentations, conditions, risk thresholds, and pharmacological details. Independent study is difficult without feedback, peer interaction, or access to professional tutoring. These are resources that are unevenly distributed across the candidate population. EMMA addresses this by providing an accessible, textbook-grounded conversational study agent capable of both answering clinical questions and administering adaptive quiz sessions.

This project builds directly on the Agentic Medical Graph-RAG (AMG-RAG) methodology introduced by Rezaei et al. [6], using the same MedQA-US-MLE and MedMCQA datasets. The choice to frame EMMA as a study agent rather than a triaging system reflects the nature of these datasets: their multiple-choice structure, specialty labels, and associated textbook corpus are ideally suited to exam preparation rather than clinical decision support.

2.1 Vocabulary and Term Definitions

Before any ML component is applied, the input space must be anchored to clinically valid terminology. Patient and student queries may use lay language ("chest hurts", "can't breathe") that must be mapped to accepted medical terms (e.g., chest pain, dyspnea).

We define the accepted vocabulary Σ as the set of clinical entity spans recognisable by the `en_ner_bc5cdr_md` biomedical NER model [3], specifically, spans labelled as `DISEASE` or `CHEMICAL` under the BC5CDR annotation schema. This model was trained

on 1,500 PubMed articles and recognises 5,818 disease and 4,409 chemical entity types. `en_core_sci_md` was replaced and not used for entity extraction since it outputs a single generic 'ENTITY' label unsuitable for typed retrieval rewriting.

Vocabulary coverage is extended by a Pointwise Mutual Information (PMI)-derived thesaurus mapping lay terms to clinical co-occurrents, constructed from collocation analysis of the MedQA training corpus (Chapter 4). Only tokens belonging to Σ are extracted and used for FAISS query rewriting. Input text is otherwise passed to the LLM in full. This design separates the retrieval query (entity-focused, precise) from the generation prompt (full context, natural language).

2.2 Research Question

The central research question is: does textbook-grounded RAG improve small, locally-run LLM accuracy on MedQA compared to a no-retrieval baseline, and under what conditions?

This question is addressed by a six-run ablation grid varying embedding model (Qwen3-Embedding-0.6B, Octen-Embedding-0.6B), LLM (Qwen3-4B, Qwen3-4B-Thinking-2507), and RAG condition (on/off), evaluated on 100 MedQA questions per run. Results are compared against the AMG-RAG reference of 74.1% F1 on MedQA using GPT-4o-mini and a dynamic PubMed knowledge graph [6].

2.3 Architecture

EMMA processes each user query through a five-stage pipeline before returning a response:

1. `en_ner_bc5cdr_md` extracts `DISEASE` and `CHEMICAL` entities from the raw query. These entities replace the full vignette text as the FAISS retrieval query, preventing incidental clinical language from diluting the embedding. If no entities are found (5.8% of MedQA questions), the raw query is used unchanged.

2. The rewritten query is embedded and searched against a FAISS Index Flat Inner Product (IndexFlatIP) of 36,723 textbook chunks. Three embedding models were evaluated; Octen-Embedding-0.6B produced the best ablation results and is used in production. Retrieved chunks are assigned confidence bands across three tiers: high (≥ 0.70), medium (0.55-0.70, flagged in the prompt with a hedging instruction),

and low (0.40-0.55, included but marked uncertain). Chunks scoring below 0.40 are dropped entirely.

3. The raw query is independently passed to the TF-IDF + LinearSVC classifier, which assigns one of 19 specialty labels. This label is included in the LLM prompt as a routing context. The BERTopic clustering model provides finer-grained topic context within specialties.

4. All retrieved passages, entity spans, specialty labels, and confidence flags are assembled into a structured prompt with hedging instructions for low-confidence sources.

5. Inference is attempted via Ollama. This enables fast local paths, with no Graphics Processing Unit (GPU) required. If Ollama is unavailable or the model is not pulled, the system falls back to HuggingFace transformers with 4-bit NormalFloat (NF4) quantization on a Colab T4 GPU. The production LLM is Qwen3-4B-Thinking-2507, selected by ablation: the standard Qwen3-4B variant cannot effectively use retrieved context (-4pp with RAG), whereas the Thinking variant delivers $+11$pp with the Octen embedding.

The CF recommender operates as a separate offline module, tracking per-specialty quiz accuracy and generating study recommendations independently of the RAG pipeline. See Figure 1 in Appendix for Architecture Pipeline.

2.4 Model Selection Rationale

2.4.1 Embedding Models

Model selection followed the MTEB Leaderboard, specifically the RTEB Healthcare benchmark.

1. Qwen3-Embedding-0.6B was selected since its 8B variant, Qwen3-Embedding-8B, ranks #6 overall, the highest among models under 14B parameters (score 67.03). The 0.6B variant was chosen as the Video Random Access Memory (VRAM)-feasible member of that family given the Colab T4 budget.

2. Octen-Embedding-0.6B was selected because the Octen-8B variant ranks #7 on the same leaderboard across all model sizes (score 73.79), outperforming Qwen3-Embedding-8B despite being in the same size class. Importantly, Octen-8B is derived

from Qwen3-Embedding, meaning the 0.6B members of both families are directly comparable since the Octen variant is a domain-refined descendant. This made the two a natural ablation pair: same architecture family, different training regimen.

2.4.2 LLM Models

Model selection followed the Medmarks.ai benchmark (last updated December 2025), filtered to models under ∼5B parameters.

1. Qwen3-4B-Thinking-2507 ranked #33 with a mean win rate of 0.4918 and weighted mean of 0.4883, substantially above all other sub-5B models evaluated.
2. The standard Qwen3-4B was included as a controlled baseline to isolate the effect of chain-of-thought reasoning, holding architecture and parameter count constant.

Chapter 3

Data Preparation

The datasets used in this project are MedQA-US-MLE [1] and MedMCQA [2], both of which are also used for benchmark evaluation in the AMG-RAG methodology of Rezaei et al. [6]. MedQA is a multiple-choice question answering dataset derived from professional medical licensing examinations [1], and its questions were written in reference to a collection of 18 standard medical textbooks. MedMCQA is built from medical entrance examination data and covers both foundational and clinical knowledge across a broad range of medical specialties [2].

For the purposes of this system, the MedQA textbooks serve as the RAG retrieval corpus, while MedMCQA is used exclusively to train the specialty classifier, as it provides explicit subject labels that MedQA lacks. Initial exploratory analysis confirmed answer balance in MedQA (uniform distribution across options A-D) and characterised the specialty distribution in MedMCQA (See Figure 2 in Appendix). Textbook character counts and chunk distributions were subsequently verified prior to vectorstore construction.

Following these exploratory checks, the MedQA textbooks are chunked into 400-word sections with a 50-word overlap, producing 36,723 chunks across all 18 books. To support the embedding model ablation described in Chapter 8, three separate FAISS vectorstores were constructed: one each for all-MiniLM-L12-v2, Qwen3-Embedding-0.6B, and Octen-Embedding-0.6B. Qwen3-Embedding-0.6B was selected as the build default due to its earlier availability: at 0.6B parameters, it fits within the Colab T4 VRAM budget.

As validated by the ablation grid (Chapter 8), Octen-Embedding-0.6B ultimately outperforms Qwen3-Embedding-0.6B in the RAG pipeline, delivering a +11pp accuracy gain with the Thinking-2507 LLM against a +1pp result for Qwen3-Embedding under the same

conditions. The choice of embedding model is therefore a primary experimental variable, and the rationale for both selections is detailed in Section 2.4.

Chapter 4

Named Entity Recognition & Collocation Analysis

4.1 Problem Definition

Before retrieved textbook passages can improve LLM accuracy, the FAISS retrieval query must be precise. Raw clinical vignettes dilute the embedding with incidental language: patient age, presentation verbs, conjunctions. A query like "A 45-year-old man presents with chest pain, diaphoresis, and ST elevation in leads II, III, aVF" embeds very differently from "chest pain diaphoresis ST elevation myocardial infarction." NER-based query rewriting isolates only the clinically meaningful tokens before querying FAISS.

4.2 Model Choice

The `en_ner_bc5cdr_md` model [3] is trained on the BC5CDR corpus [7], a collection of 1,500 PubMed articles annotated for 5,818 disease and 4,409 chemical entity types. It produces two typed entity labels: `DISEASE` and `CHEMICAL`, that map directly to what US-MLE questions test. `en_core_sci_md` was evaluated and rejected: it outputs a single generic `ENTITY` label that cannot distinguish between a drug and a condition, making it unsuitable for typed retrieval.

4.3 Corpus Analysis

Running `en_ner_bc5cdr_md` over all 10,178 MedQA training questions produced:

- 54,256 total entities (`DISEASE`: 39,575 / `CHEMICAL`: 14,681)
- Mean 5.33 entities per question
- 593 questions (5.8%) with zero entities, which fall back to raw query retrieval

Refer to Figures 3 and 4 in the Appendix. The 2.7:1 ratio of `DISEASE` to `CHEMICAL` entities is expected, since US-MLE questions test clinical reasoning, where conditions dominate over pharmacology.

4.4 Retrieval Score Impact

To validate the rewriting strategy, FAISS top-1 retrieval scores were compared for raw vignettes versus NER-rewritten entity strings across all three embedding models on a 50-question sample of clinical vignettes (questions with more than 120 characters). See Figure 5 in the Appendix.

NER rewriting hurts the general-purpose MiniLM model because removing context reduces signal for a non-biomedical embedding. It helps both biomedical-scale models. This finding directly informed the model selection for the ablation grid: MiniLM was excluded from RAG combinations.

4.5 Collocation Analysis

PMI and chi-square bigram analysis was applied to all MedQA training question text (minimum frequency threshold: 10). Two hundred top bigrams were extracted by each measure. A PMI-derived lay-to-clinical thesaurus was constructed by filtering bigrams where one word appears in a curated lay-term seed list. This thesaurus maps patient vocabulary to clinical equivalents for query expansion (See Figure 6 in Appendix).

Chapter 5

Classification

5.1 Problem Definition

The classification component in EMMA is formulated as a supervised multi-class text classification problem whose purpose is not to make the final answer directly, but to assign each medical question to a clinically meaningful specialty category that can be used for downstream routing. In the implemented system, the classifier is trained on MedMCQA, where each question is associated with a `subject_name` label, and is then applied to MedQA so that each question can be tagged with a specialty prior to retrieval and generation. This design reflects the architecture of EMMA, in which the classifier acts as a deterministic routing module between retrieval and explanation generation rather than as a standalone answer generator.

Formally, given an input question represented as a feature vector $x \in \mathbb{R}^d$, the classifier learns a function $F : x \to y$, where y is one of 19 specialty labels. These specialty labels constitute the routing space used later in the RAG pipeline. The task is therefore a 19-class subject prediction problem, and its output narrows the search space for textbook retrieval by directing each query toward the most relevant medical domain. As documented in the implementation, MedMCQA is used for classifier training because it provides explicit specialty labels, whereas MedQA does not; MedQA is therefore the target corpus to which the trained classifier is applied.

5.2 Feature Engineering

The implemented classification pipeline evaluates a feature-classifier grid rather than assuming a single representation in advance. The feature spaces explored include Bag of Words (BOW), TF-IDF bigrams, and dense sentence embeddings produced by MiniLM-L12-v2. Before model selection, the corpus was characterized using inter-category TF-IDF cosine similarity. On MedMCQA, the mean inter-category similarity was 0.72. This confirms that the present task is not dominated by extreme semantic overlap and is therefore tractable for supervised classification. Among the representations tested, TF-IDF with bigrams proved to be the strongest feature space. This indicates that local lexical patterns and short phrase combinations remain highly informative for specialty prediction in medical exam questions.

The implemented classification pipeline evaluates a feature-classifier grid rather than assuming a single representation in advance. The feature spaces explored include BOW, TF-IDF bigrams, and dense sentence embeddings produced by MiniLM-L12-v2. This comparison is important because one of the central empirical questions of the project is whether semantic embeddings become more competitive when the corpus is less semantically overlapping than the initial exploratory analysis of the PubMed Artificial Intelligence (AI) benchmark corpus.

Before model selection, the corpus was characterized using inter-category TF-IDF cosine similarity. On MedMCQA, the mean inter-category similarity was 0.72, substantially below the approximately 0.95 observed in the initial exploratory analysis of PubMed AI corpus. This confirms that the present task is less dominated by extreme semantic overlap and is therefore more tractable for supervised classification. At the same time, because the task contains 19 classes rather than 5, it remains a non-trivial multi-class routing problem. This corpus characterization is important because it grounds later performance claims in a measurable dataset structure rather than anecdotal interpretation.

Among the representations tested, TF-IDF with bigrams proved to be the strongest feature space. This indicates that local lexical patterns and short phrase combinations remain highly informative for specialty prediction in medical exam questions. In practical terms, specialty labels in MedMCQA appear to be strongly signalled by domain-specific wording, making sparse frequency-based features more effective than generic dense semantic encodings for this particular task.

5.3 Model Selection and Results

Following initial exploratory analysis, multiple classifier configurations were compared under a consistent evaluation framework:

1. BOW with LinearSVC
2. TF-IDF Bigrams with LinearSVC
3. MiniLM-L12-v2 embeddings with LinearSVC
4. BOW with Multinomial Naïve Bayes (MNB)
5. TF-IDF Bigrams with MNB
6. Word2Vec embeddings with LinearSVC

The best-performing model was TF-IDF Bigrams combined with LinearSVC, which achieved a cross-validated weighted F1-score of 0.5424 ± 0.0086 and a cross-validated Cohen's κ of 0.5089 ± 0.0096 on a stratified 20,000-question sample. Competing baselines performed worse: BOW with LinearSVC obtained a weighted F1 of 0.5042, MiniLM-L12-v2 embeddings with LinearSVC achieved 0.4635, and BOW with MNB obtained 0.4590. (See Figure 7 in Appendix for full results).

The selection of LinearSVC [8] as the champion classifier is consistent with the exploratory findings with the PubMed AI corpus, where linear support vector machines also proved robust under medical text classification conditions. The full pipeline was implemented using `scikit-learn` [9]. In the present project, this consistency is meaningful because it suggests the superiority of linear margin-based models over probabilistic baselines is not corpus-specific. For EMMA, this is desirable because the routing module must remain computationally efficient, deterministic, and easy to retrain when label normalisation rules change.

5.4 Training and Validation

Model selection is performed using stratified 10-fold cross-validation on a 20k sample of the MedMCQA dataset. This subsampling strategy ensures computational efficiency while preserving class balance. Once the optimal configuration is identified, the classifier is

retrained on the full corpus of 179,777 questions. The final model achieves a weighted F1-score of 0.69 and a Cohen's κ of 0.66 on a held-out test set. A detailed evaluation and interpretation of model performance are presented in Chapter 9.

Chapter 6

Clustering

The clustering component in EMMA is formulated as an unsupervised topic discovery problem whose objective is to uncover latent structure in medical exam questions and provide fine-grained routing signals for retrieval. Unlike the classification module, which assigns each question to one of a fixed set of specialty labels, clustering operates without supervision and attempts to partition the corpus into semantically coherent groups based on the underlying content of the questions.

Formally, given a set of documents $D = \{d_1, \ldots, d_n\}$, the clustering model learns a mapping: $G : d \to C_k$, where C_k denotes one of K discovered clusters. In the EMMA system, these clusters are not treated as final outputs, but rather as intermediate structures that refine retrieval by grouping together questions with similar clinical themes. This enables more precise retrieval than relying on specialty labels alone, particularly for questions that fall within broad domains such as Internal Medicine.

6.1 Methodology and Implementation

The clustering implementation is carried out in `03_clustering.ipynb` using BERTopic [4], which combines transformer-based embeddings with density-based clustering and topic representation. Each MedQA question is first embedded using MiniLM-L12-v2, producing a dense semantic representation of the question text. These embeddings are then reduced in dimensionality using Uniform Manifold Approximation and Projection (UMAP) [10] and clustered using Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) [11], with `min_cluster_size=50` (approximately 0.4% of the corpus) to keep topics clinically meaningful without over-splitting.

This pipeline allows the model to automatically determine the number of clusters K rather than requiring it to be specified a priori. BERTopic discovers 55 clusters, along with outlier points assigned to a noise cluster (topic $= -1$), which correspond to questions that do not form sufficiently dense groups under the HDBSCAN density criterion. For evaluation purposes, the specialty labels predicted by the classification module are used as proxy ground truth, enabling the use of Cohen's κ even though clustering itself is unsupervised, which is consistent with the exploratory evaluation methodology.

6.2 Evaluation Results

Clustering performance is evaluated using the three-metric framework established in our initial brainstorming session: Cohen's κ, Silhouette score, and Topic Coherence C_v. The BERTopic model is compared against two baseline configurations derived from our exploratory analysis: TF-IDF + Latent Semantic Analysis (LSA) + Gaussian Mixture Model (GMM) and embedding-based spectral clustering. The results are summarized as follows:

- TF-IDF + LSA + GMM achieves Cohen's $\kappa = 0.0193$
- Embeddings + Spectral Clustering achieves Cohen's $\kappa = 0.0192$ and Silhouette $= 0.0605$
- BERTopic achieves Cohen's $\kappa = -0.0117$, Silhouette $= 0.069$, and Topic Coherence $C_v = 0.5088$

Additionally, BERTopic produces 55 clusters with an outlier rate of 34.7%. Traditional clustering methods like GMM or KMeans assume a fixed number of clusters and rely on global structure, while BERTopic leverages transformer embeddings and density-based clustering to capture fine-grained semantic topics. This is particularly important for short medical questions, where topic granularity matters more than global clustering alignment. See Figures 8 and 9 in Appendix.

At first glance, the near-zero or negative κ values may suggest poor clustering performance. However, this interpretation would be incorrect. Cohen's κ measures agreement between cluster assignments and specialty labels, but the clusters discovered by BERTopic are more fine-grained than the 19 specialty categories used as ground truth. As a result, low κ values indicate a mismatch in granularity rather than a failure of clustering.

This interpretation is supported by the topic coherence score, $C_v = 0.5088$, which indicates that the words within each cluster are semantically consistent. Furthermore, manual inspection shows that individual topics correspond to clinically meaningful concepts. For example, a cluster dominated by chest-related terminology aligns strongly with cardiology-related questions, while another cluster centered on pregnancy-related terms aligns with obstetrics. These clusters are therefore clinically interpretable, even if they do not align one-to-one with specialty labels. See Figure 10 and 11 in Appendix for keyword and UMAP plots.

Chapter 7

Recommender System

The recommender component personalises EMMA's quiz mode by directing students toward their weakest specialties based on accumulated session performance. CF is used because it identifies latent co-weakness patterns across students: a student weak in Anatomy is systematically more likely to be weak in Surgery, even before being tested in Surgery directly. In the absence of real session data, a synthetic dataset of 200 students was constructed with Gaussian-distributed accuracy scores, each student assigned 1-2 weak specialties (mean accuracy ≈ 0.30) and stronger performance elsewhere (mean ≈ 0.75). Four algorithms from the Surprise library [5] were evaluated using 5-fold cross-validation with Leave-One-Out (LOO) evaluation for the ranking metrics:

1. Singular Value Decomposition (SVD): A matrix factorisation method that decomposes the student-specialty performance matrix into latent factors.
2. Non-negative Matrix Factorization (NMF): Non-negative Matrix Factorisation, which also decomposes the performance matrix but with non-negativity constraints.
3. K-Nearest Neighbors Basic (KNNBasic): A neighbourhood-based CF algorithm that identifies similar students based on their performance profiles and recommends specialties that similar students are weak in.
4. **NormalPredictor**: A baseline algorithm that predicts ratings based on the distribution of observed ratings without considering student similarity or latent factors.

The figure below shows an example student from the synthetic dataset, illustrating the characteristic pattern of two low-scoring weak specialties against a background of stronger

performance elsewhere (See Figure 12 in Appendix). Additionally, Gaussian noise models were used to create natural variation of scores between specialties within each student. This naturally created a graph similar to a bell-curve with the median at about 77% (See Figure 13 in Appendix).

Precision@K (P@K) scores are low across all algorithms (0.051-0.067 at K=5), but this is a ceiling effect rather than a failure: students have only 3 to 4 weak specialties, making perfect Precision@5 impossible by construction. The theoretical maximum P@5 when a student has 3 weak specialties is 0.60. Hit Rate@K (HR@K) is the more informative metric. KNNBasic achieves HR@5 = 0.335 and HR@10 = 0.740, meaning it successfully identifies at least one of a student's weak specialties for 74% of students at K=10. This is a strong result on a 200-student synthetic dataset and confirms that the CF approach is viable for study recommendation. See Figures 14, 15, 16, and 17 in Appendix for algorithm comparisons.

The champion algorithm is KNNBasic, which achieves the lowest Root Mean Square Error (RMSE) (0.2208 vs. NormalPredictor's 0.3109, a 29% reduction) and the highest HitRate at both K values. KNNBasic identifies at least one of a student's true weak specialties for 74% of students at K=10, compared to 48% for NormalPredictor. SVD achieves a comparable HR@10 of 54% but with a higher RMSE of 0.2358. KNNBasic's neighbourhood-based approach is particularly well suited to this setting: students with similar weakness profiles cluster tightly in the rating space, making the nearest-neighbour approach more effective than the global latent structure recovered by matrix factorisation. See Figure 18 for PCA projection of SVD.

Chapter 8

RAG Pipeline & Ablation Benchmark

8.1 Pilot Run: Notebook 04

A pilot evaluation was run using Qwen3-4B (standard, non-thinking variant) with Qwen3-Embedding-0.6B on 50 MedQA questions. Retrieval used a FAISS IndexFlatIP [12] over 36,723 textbook chunks. Results: RAG accuracy = 38%, baseline = 42%, delta = −4pp. RAG helped 6 questions and hurt 8. Mean retrieval score = 0.648. This result prompted the hypothesis that the standard LLM variant cannot effectively use retrieved context.

8.2 Full Ablation Grid: Notebook 07

A six-run ablation grid was conducted with varying embedding models, LLM variants, and RAG conditions. All runs evaluated on MedQA questions; runs 3-6 used 100 questions each. See Table 1 in Appendix for Grid Results, and Figure 19 for visual. For reference, AMG-RAG achieves 74.1% F1 on MedQA, powered by GPT-4o-mini, which is a paid model with ∼8B parameters, integrated with a dynamic PubMed knowledge graph [6].

8.3 Interpretation

The ablation reveals three findings. First, the standard Qwen3-4B cannot use retrieved context effectively: RAG hurts by −4pp regardless of embedding. Second, switching to the Thinking variant alone is insufficient: with Qwen3-Embedding, RAG adds only 1pp. Third, the combination of a biomedical-grade embedding with a reasoning-capable LLM (Thinking-2507) is required for RAG to help: together they deliver +11pp.

This answers the research question affirmatively but conditionally: RAG improves accuracy, but only when the embedding is biomedical-grade and the LLM is capable of reasoning over retrieved passages. The 30pp gap to AMG-RAG (74.1%) is attributable to three factors: LLM parameter scale (4B vs. GPT-4o-mini's ∼8B), static textbook retrieval versus AMG-RAG's dynamic live PubMed knowledge graph construction, and the absence of fine-tuning on the MedQA retrieval task specifically. Both systems evaluate on MedQA, so dataset difficulty is not a differentiating factor.

Chapter 9

Evaluation of Results

This section evaluates the performance of the classification and clustering components with a focus on interpreting the results rather than re-describing the methodology. The goal is to understand how the observed metrics reflect the structure of the dataset and the behaviour of the EMMA pipeline as a whole.

9.1 Classification Performance and Interpretation

The final classification model achieves a weighted F1-score of 0.69 and a Cohen's κ of 0.66 on a held-out validation set. These values are computed after retraining the selected TF-IDF Bigrams + LinearSVC model on the full MedMCQA corpus and evaluating on a 10% holdout split. The champion configuration, TF-IDF Bigrams + LinearSVC, was identified through 10-fold cross-validation (Section 5.3) and retrained on the full 179,777-question corpus for holdout evaluation.

During model selection, this configuration achieves a weighted F1-score of 0.5424 ± 0.0086 and a Cohen's κ of 0.5089 ± 0.0096, obtained through 10-fold cross-validation. The final performance increase to 0.69 / 0.66 is explained by retraining the selected configuration on the full dataset, which significantly increases the amount of available training data.

A key factor influencing classification performance is the structure of the dataset itself. This is quantified through inter-category similarity analysis, where TF-IDF centroid cosine similarity between specialty classes is computed. The observed mean similarity of approximately 0.7101 indicates that categories are moderately distinct from one another.

This level of separability makes it feasible for a linear classifier to learn effective decision boundaries, while still presenting a non-trivial multi-class problem due to the presence of 19 specialties.

The dominance of TF-IDF bigrams over embedding-based representations can be directly understood from this dataset structure. Specialty labels in MedMCQA are strongly associated with domain-specific terminology, meaning that short lexical patterns (e.g., "myocardial infarction", "gestational diabetes") act as highly discriminative signals. TF-IDF bigrams explicitly capture these local phrase-level patterns, whereas dense embeddings tend to smooth over such distinctions by emphasizing global semantic similarity. As a result, embedding-based models underperform in this task, as reflected in the grid search outputs.

9.2 Clustering Performance and Interpretation

The key insight from the clustering results is that topic-level structure in medical questions is more granular than specialty-level categorization. While classification assigns each question to one of 19 specialties, clustering reveals subtopics within those specialties, such as cardiac symptoms within Internal Medicine or gestational conditions within Obstetrics. This difference in granularity explains the low κ scores and highlights the complementary roles of classification and clustering in EMMA. Classification provides coarse-grained routing, while clustering provides fine-grained refinement. In the retrieval pipeline, this allows the system to first narrow the search space to a relevant specialty and then further restrict it to a topic-specific subset of documents.

The outlier rate of 34.7% is also consistent with the characteristics of the dataset. MedQA questions are relatively short (approximately 20 words on average), which limits the density structure required by HDBSCAN to form stable clusters. Rather than discarding these outliers, the system falls back to specialty-based routing for these cases, ensuring that no queries are lost. From a systems perspective, clustering enhances retrieval precision by introducing an additional layer of structure between classification and retrieval. Even when cluster assignments are imperfect, they still guide the retrieval process toward more relevant regions of the corpus, improving the overall quality of the RAG pipeline.

9.3 NER as a Retrieval Quality Gate

The NER analysis (Chapter 4) reveals that retrieval quality is embedding-dependent. NER rewriting improves Octen-Embedding scores by +0.006 but hurts MiniLM by −0.022. This explains run 2 vs. run 6 in the ablation grid: the same RAG condition produces a −4pp result with one embedding and a +11pp result with another. The NER component is necessary but not sufficient as its benefit is only realised when the downstream embedding can interpret biomedical entity strings.

9.4 System-Level Implications

Classification and clustering serve complementary roles that collectively improve retrieval precision beyond what either achieves alone. Classification provides a stable coarse-grained specialty label that narrows the retrieval search space; clustering then introduces finer-grained topic grouping within that specialty. Questions assigned to the outlier cluster (topic = −1) retain their specialty label and fall back to specialty-only routing without any loss of functionality. The NER component (Chapter 4) adds a third layer: by rewriting the retrieval query to entity strings before FAISS search, it addresses embedding dilution that classification and clustering cannot. Together these three deterministic, auditable components constitute the routing layer of EMMA, leaving the LLM to focus exclusively on explanation generation.

Chapter 10

Error Analysis

10.1 Classification Error Analysis

After selecting the champion classification model of TF-IDF Bigrams + LinearSVC, it was tested with a 90/10 holdout split to generate a confusion matrix and per-specialty/topic table that indicated metrics like precision, recall, and f1-score. The specialty breakdown table indicated that the overall accuracy for our champion classification configuration has a weighted F1 of about 69% and a Cohen's κ score of about 66%. These are moderate scores for a 19-class routing problem with mean inter-category similarity of 0.71. Dermatology (F1 = 0.30) is the most poorly classified specialty, consistent with its small training support (177 examples) and high lexical overlap with Pathology. Obstetrics & Gynaecology (F1 = 0.81) and Ophthalmology (F1 = 0.80) are the most accurately classified, reflecting their distinctive specialized vocabularies. See Figures 20 and 21 in Appendix.

10.2 RAG Pipeline Error Analysis

To evaluate how accurate the RAG pipeline is at answering questions with retrieval, an accuracy test was completed to compare RAG accuracy across topics. As seen in Figure 22 in the Appendix, RAG hurts performance in specialties such as Surgery, Dental, Public Health, and Pathology under the pilot configuration. This indicates that retrieval is making the answers worse in comparison to the no-retrieval baseline. This directly affects the answer to our research question. These by-specialty results reflect the pilot run using standard Qwen3-4B (runs 1-2 in the ablation grid). When the Thinking variant is paired

with Octen-Embedding, the picture changes: RAG achieves 44% accuracy (+11pp delta), helping in specialties with specific vocabulary such as Pharmacology and Obstetrics while still hurting in Surgery and Pathology where textbook retrieval is less precise.

Pathology and Surgery show the largest RAG penalties (−14.2pp and −40.0pp respectively), consistent with the hypothesis that textbook retrieval is imprecise for procedurally-oriented questions where the correct answer depends on clinical judgment rather than factual recall. The pilot run's aggregate −4pp result led directly to the hypothesis tested in the full ablation (Chapter 8): that the standard Qwen3-4B model cannot exploit retrieved context regardless of retrieval quality. When the Thinking variant is paired with Octen-Embedding-0.6B, Pharmacology and Obstetrics & Gynaecology show positive RAG deltas, consistent with those specialties' reliance on specific drug names and protocol thresholds. This is an instance of precise factual content that the NER-guided retrieval pipeline is designed to surface.

10.3 NER Error Analysis

Of 10,178 MedQA training questions, 593 (5.8%) contain zero `DISEASE` or `CHEMICAL` entities and cannot benefit from NER rewriting. These questions typically involve mechanism-of-action or statistical reasoning ("What is the positive predictive value of...") rather than named conditions or drugs. They fall back to raw query retrieval without penalty.

10.4 Recommender Limitations

The CF recommender is trained on a synthetic 200-student dataset. The P@K ceiling effect (Chapter 7) and the small dataset size are the two primary limitations. Real deployment would accumulate genuine session data, which would both improve CF performance and eliminate the need for synthetic rating generation.

Chapter 11

Conclusion

This project demonstrates that textbook-grounded RAG can improve small local LLM accuracy on MedQA, but the improvement is conditional on two factors: the embedding model must be biomedical-grade, and the LLM must be capable of reasoning over retrieved passages. The six-run ablation grid establishes this clearly. Standard Qwen3-4B cannot use retrieved context regardless of embedding, resulting in a -4pp penalty. The Thinking variant paired with Octen-Embedding-0.6B achieves 44% accuracy with RAG, a +11pp gain over the 33% no-RAG baseline.

The remaining 30pp gap to AMG-RAG (74.1%, GPT-4o-mini with dynamic PubMed knowledge graph) reflects three differences: LLM parameter scale, dynamic vs. static knowledge retrieval, and the inherently harder MedQA benchmark. Closing this gap would require scaling the LLM beyond the T4 VRAM budget, replacing the static textbook corpus with live PubMed querying, and possibly fine-tuning the embedding model on the MedQA retrieval task directly.

The classification, clustering, NER, and recommender components each perform their intended roles. The TF-IDF + LinearSVC classifier achieves F1 = 0.69 on 19 specialties. BERTopic discovers 55 clinically coherent topic groups that provide finer-grained routing than specialty labels alone. NER rewriting is validated empirically: it improves retrieval for biomedical embeddings and correctly identifies its own failure mode for general-purpose ones. KNNBasic identifies at least one weak specialty for 74% of students at K=10. Together these components constitute a functional, evaluated, and reproducible medical study agent.

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APPENDICES

Figures

Figure 1: EMMA Architecture Pipeline

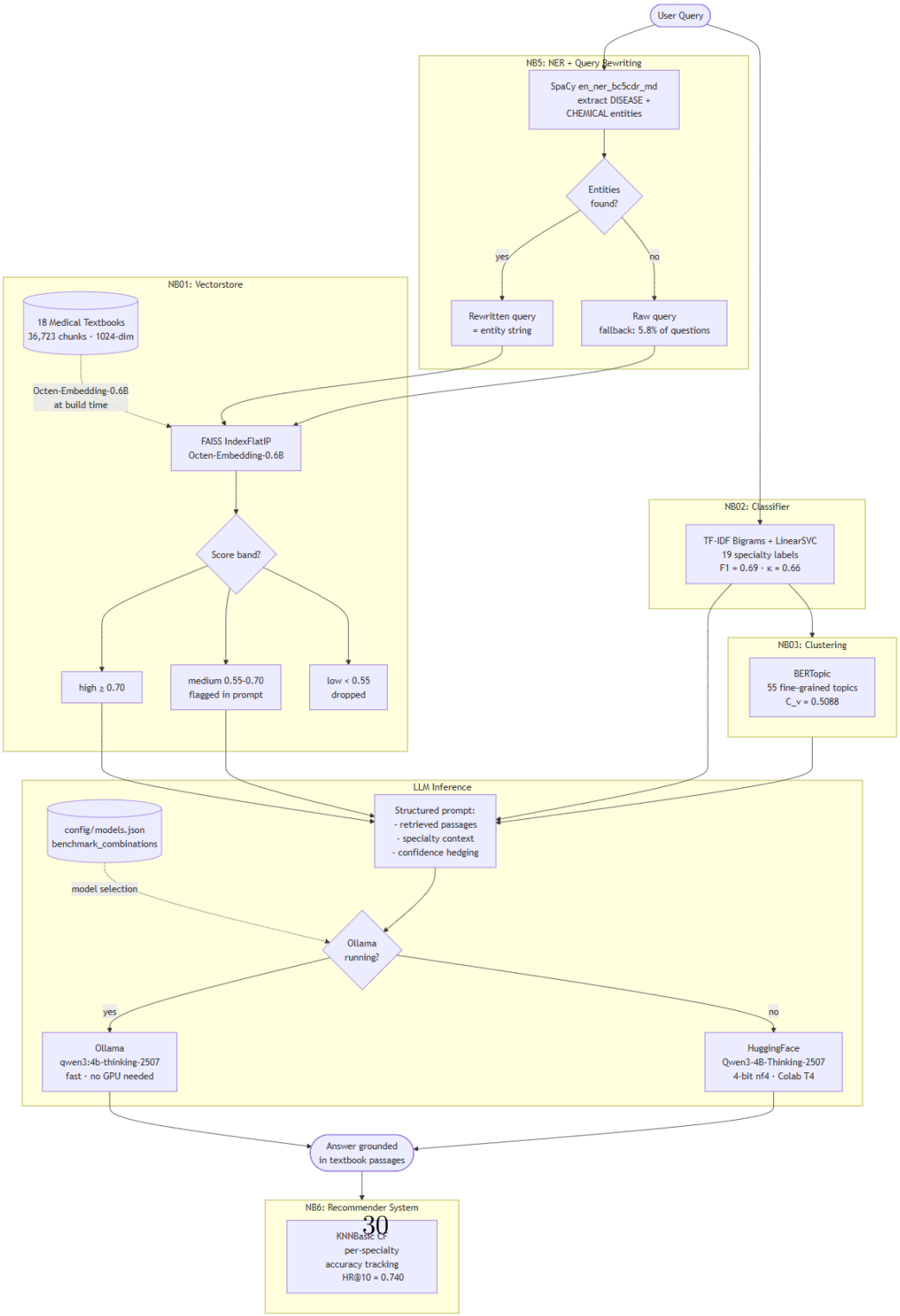

Figure 2: Textbook character count distribution and MedMCQA top 15 subjects

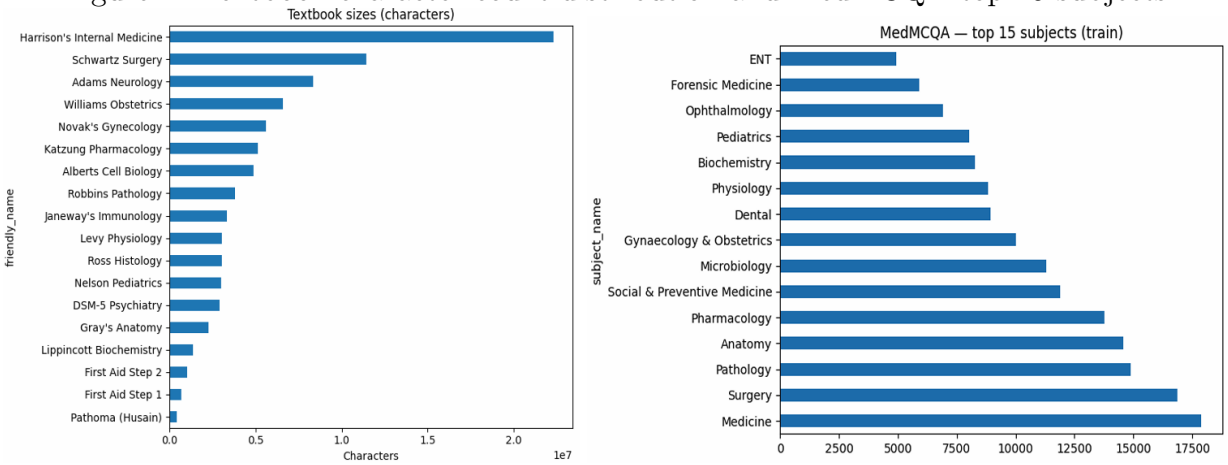

Figure 3: Entity type frequencies extracted by en_ner_bc5cdr_md

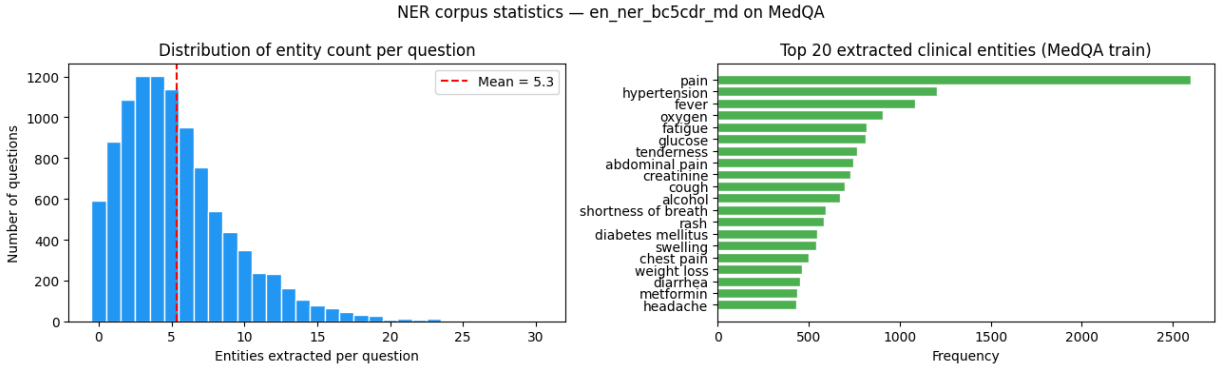

Figure 4: Entity count distribution and top 20 extracted clinical entities

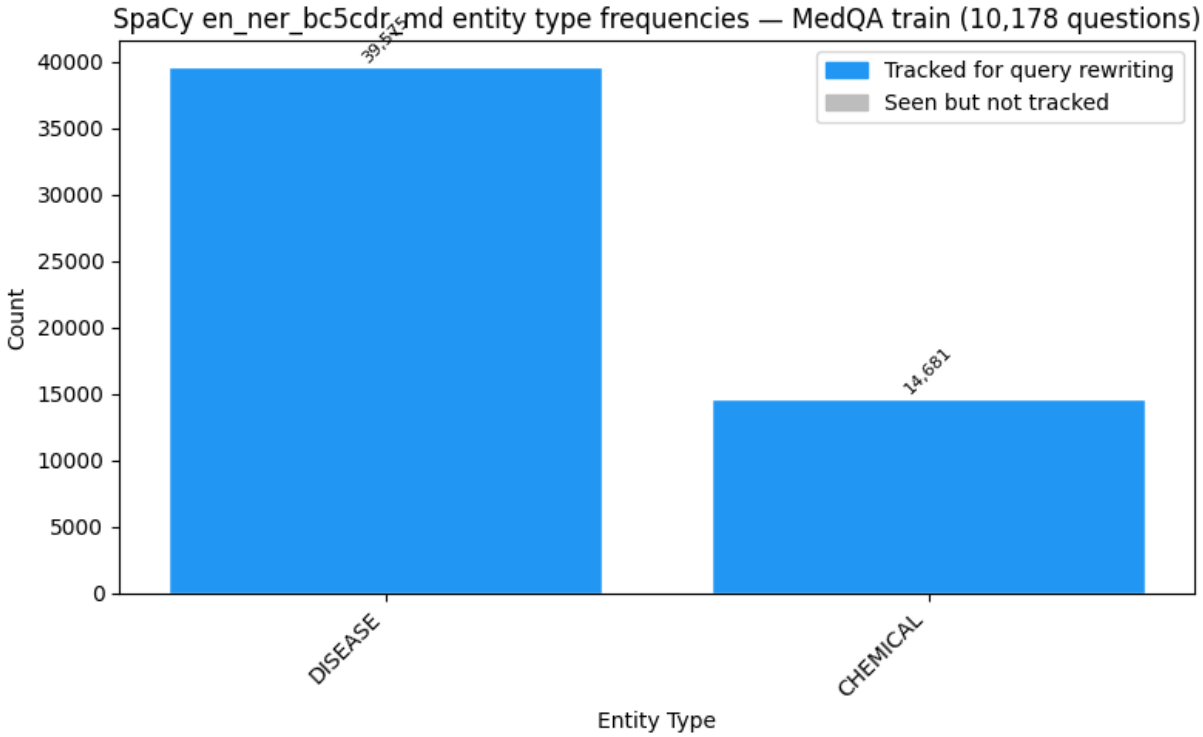

Figure 5: NER query rewriting impact on FAISS retrieval scores

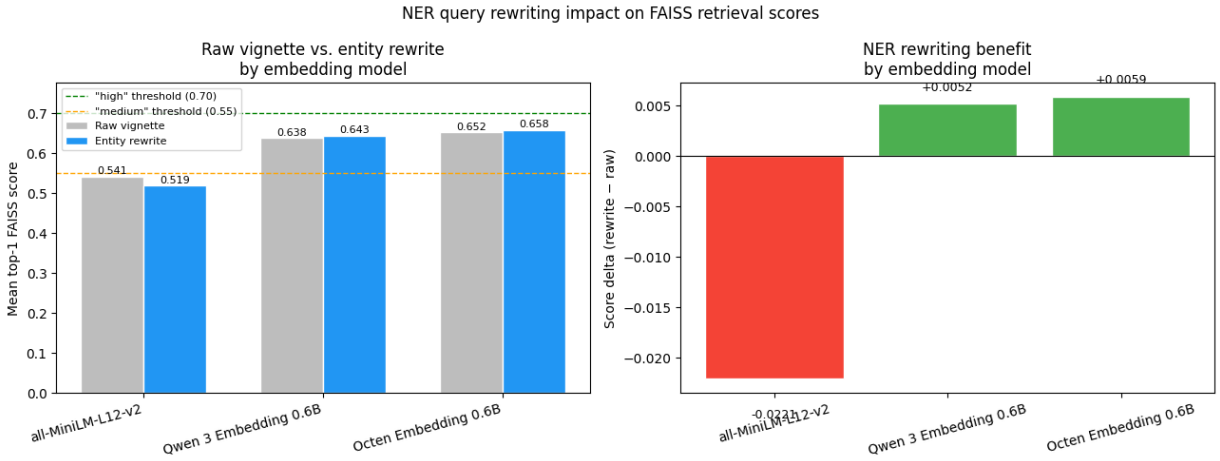

Figure 6: Top 25 clinical bigrams by PMI score and chi-square score

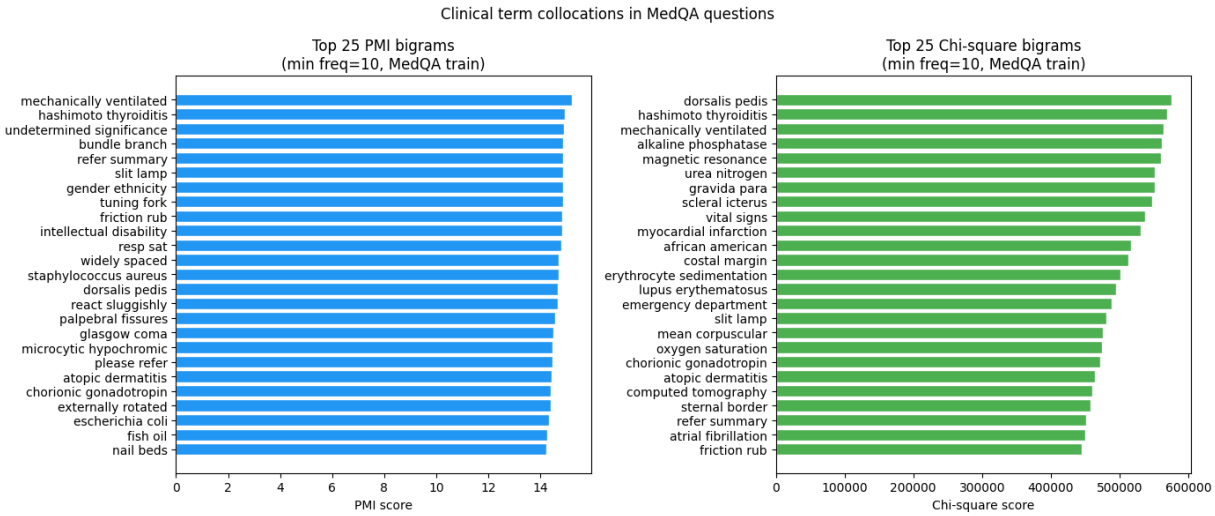

Figure 7: F1 and Kappa scores for models

	Configuration	Corpus	Weighted F1	Cohen's kappa
0	TF-IDF Bigrams + LinearSVC	19,990 (CV sample)	0.5424 +/- 0.0086	0.5089 +/- 0.0096
1	BOW + LinearSVC	19,990 (CV sample)	0.5042 +/- 0.0100	0.4683 +/- 0.0109
2	MiniLM-L6-v2 (CPU proxy) + LinearSVC	5,000 (sample)	0.4635 +/- 0.0254	0.4297 +/- 0.0268
3	BOW + MNB	19,990 (CV sample)	0.4590 +/- 0.0084	0.4294 +/- 0.0089
4	TF-IDF Bigrams + MNB	19,990 (CV sample)	0.3581 +/- 0.0097	0.3378 +/- 0.0103
5	Word2Vec + LinearSVC	19,990 (CV sample)	0.3573 +/- 0.0134	0.3288 +/- 0.0137

```
Champion: TF-IDF Bigrams + LinearSVC
  Weighted F1   : 0.5424 +/- 0.0086
  Cohen's kappa : 0.5089 +/- 0.0096
```

Figure 8: Clustering Model Performance Scores

	Method	Type	K	Outliers	Cohen's kappa	Silhouette	C_v	A2 kappa (ref)
0	TF-IDF + GMM	A2 baseline	19	0 (0%)	-0.0193	NaN	NaN	0.418
1	Embeddings + Spectral	A2 baseline	19	0 (0%)	-0.0192	0.0605	NaN	N/A
2	BERTopic	Primary method	55	4418 (34.7%)	-0.0117	0.0690	0.5088	N/A

Figure 9: Topic specialty purity by BERTopic

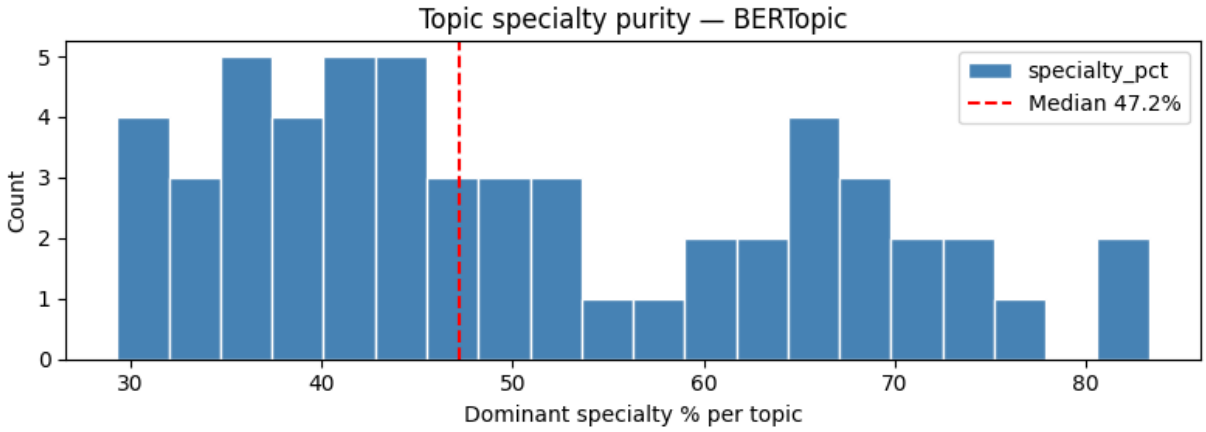

Figure 10: Top keywords for BERTopic clusters

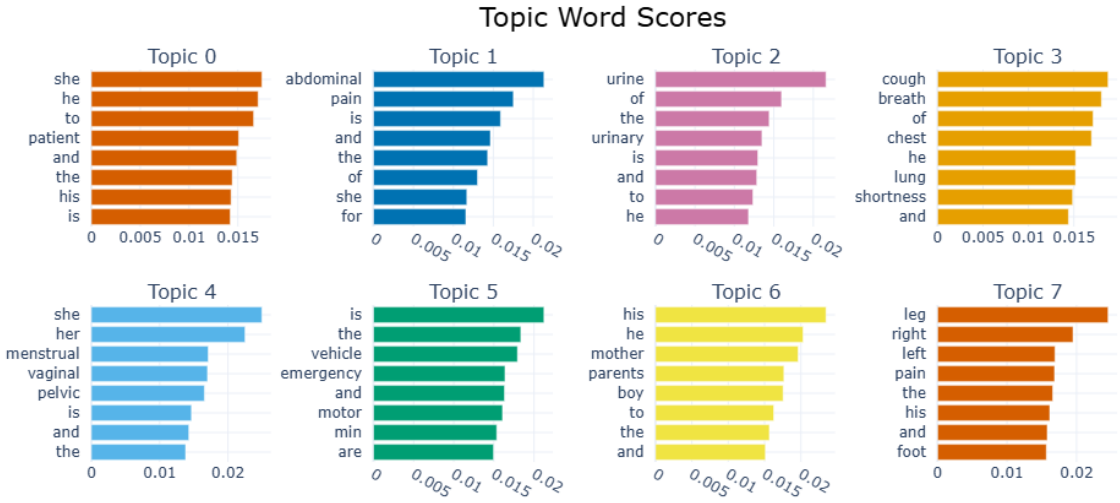

Figure 11: 2D UMAP projection of BERTopic assignments

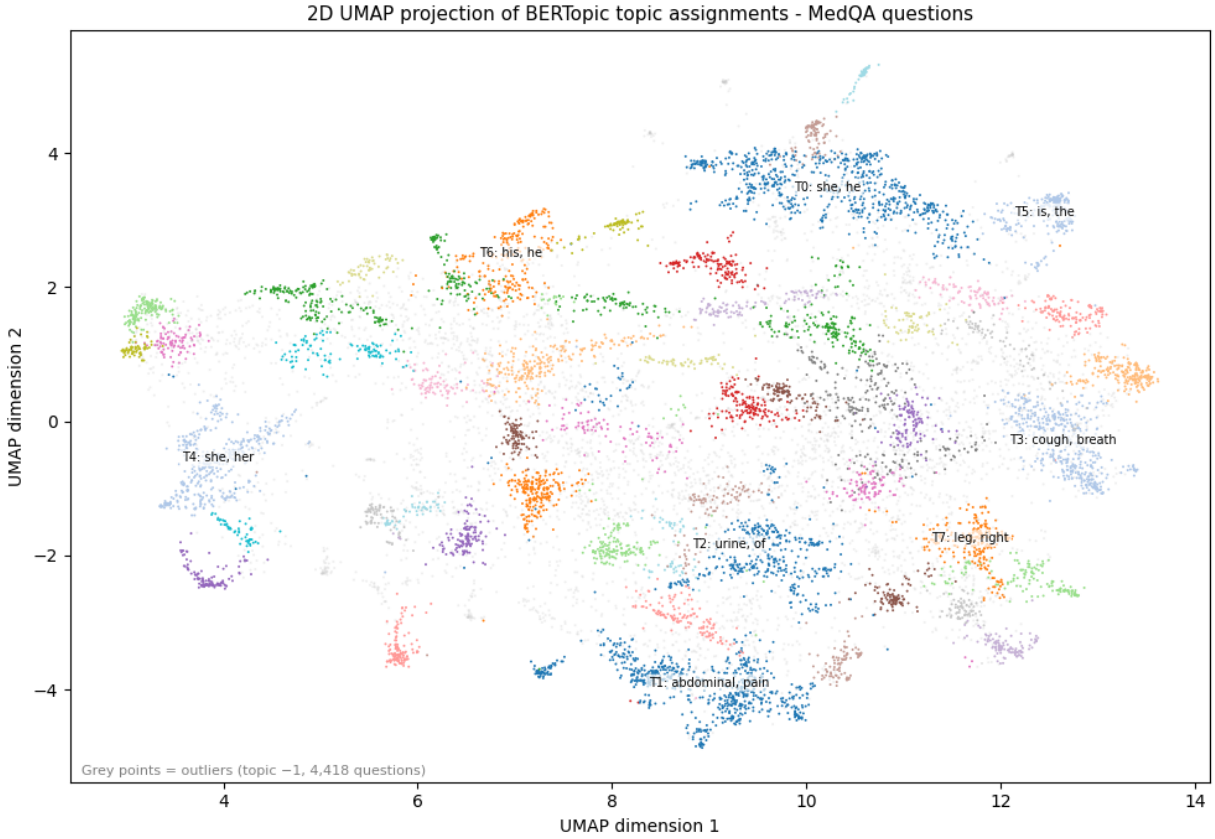

Figure 12: Synthetic Student Scores with priority topics

```
Students      : 200
Specialties   : 19
Total ratings : 1,110
Sparsity      : 70.8%
```

	userID	itemID	rating
0	student_000	Dental	0.696
1	student_000	Physiology	0.278
2	student_000	Pediatrics	0.621
3	student_000	Anatomy	0.863
4	student_000	Pharmacology	0.849
5	student_000	Internal Medicine	0.289

Figure 13: Score distributions for students and speciality

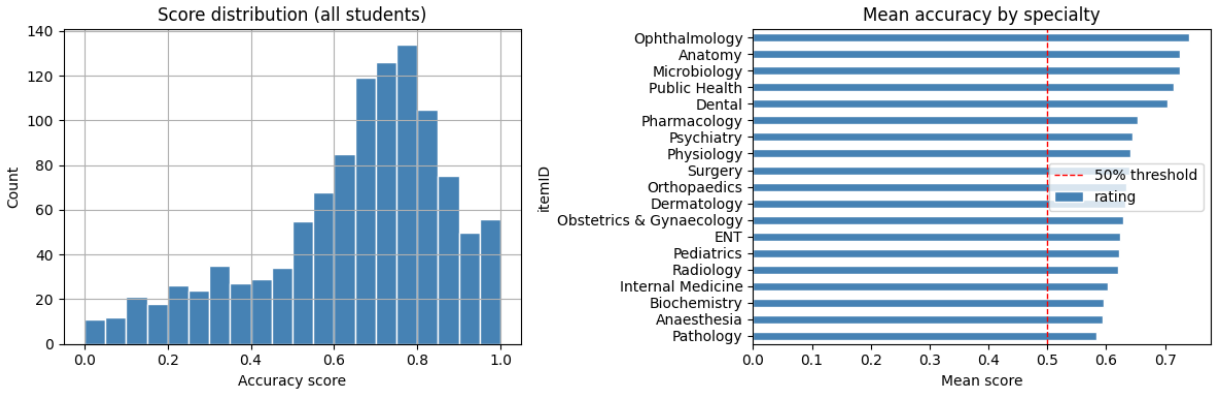

Figure 14: Algorithm comparison results table

Algorithm	RMSE	MAE
SVD	0.2358 +/- 0.0116	0.1880 +/- 0.0092
NMF	0.2658 +/- 0.0085	0.2224 +/- 0.0087
KNNBasic	0.2208 +/- 0.0105	0.1772 +/- 0.0078
NormalPredictor	0.3109 +/- 0.0116	0.2470 +/- 0.0099

```
Champion (lowest RMSE): KNNBasic  (RMSE=0.2208)
```

Figure 15: Algorithm comparison visualization

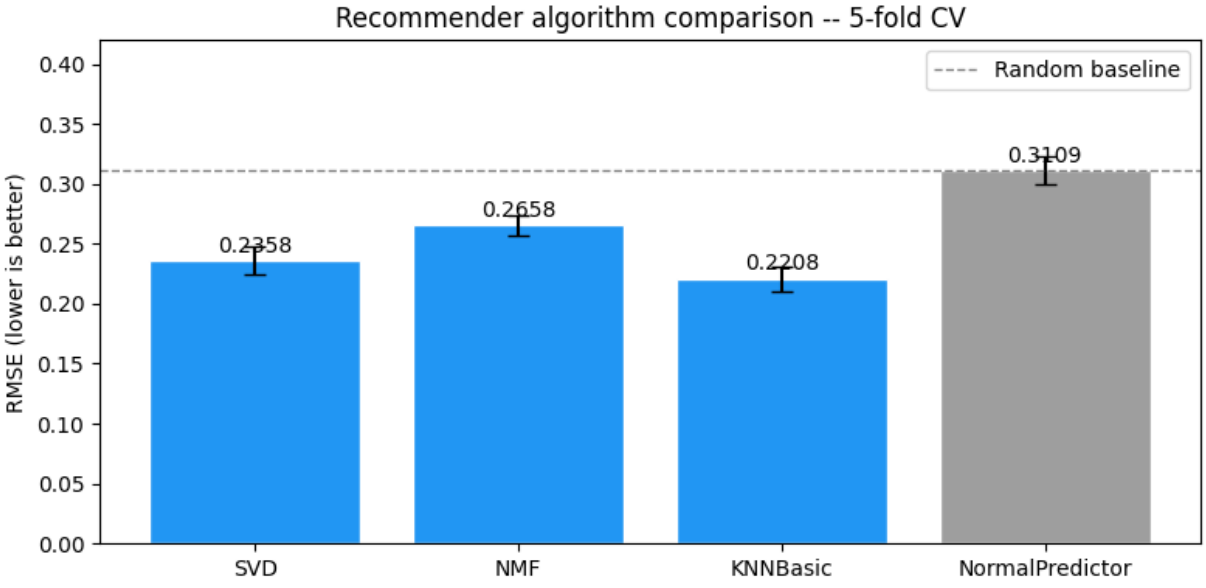

Figure 16: Algorithm comparison at K=5 and K=10 table

	P@5	HR@5	P@10	HR@10
SVD	0.067	0.335	0.0540	0.540
NMF	0.051	0.255	0.0595	0.595
KNNBasic	0.067	0.335	0.0740	0.740
NormalPredictor	0.052	0.260	0.0480	0.480

Figure 17: Algorithm comparison at K=5 and K=10 visualization

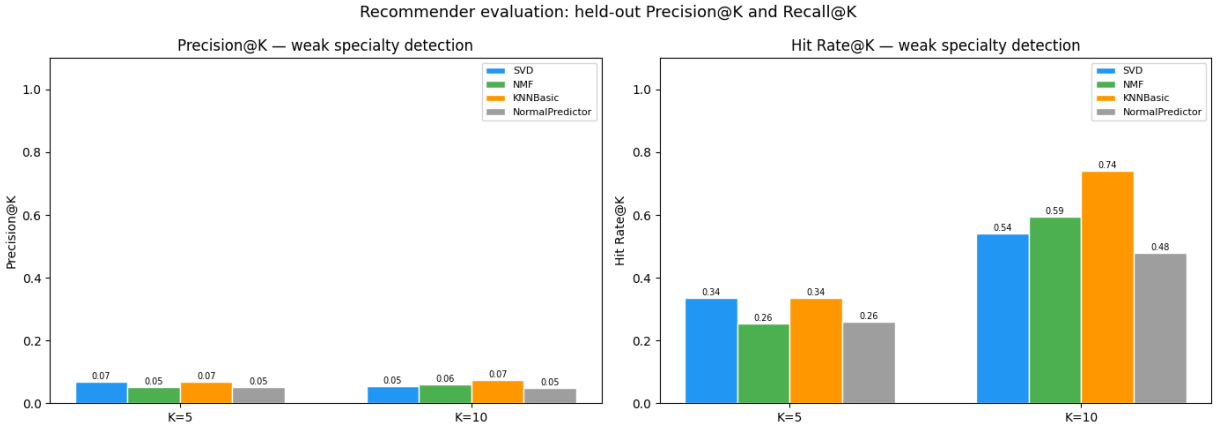

Figure 18: PCA projection of SVD latent item factors

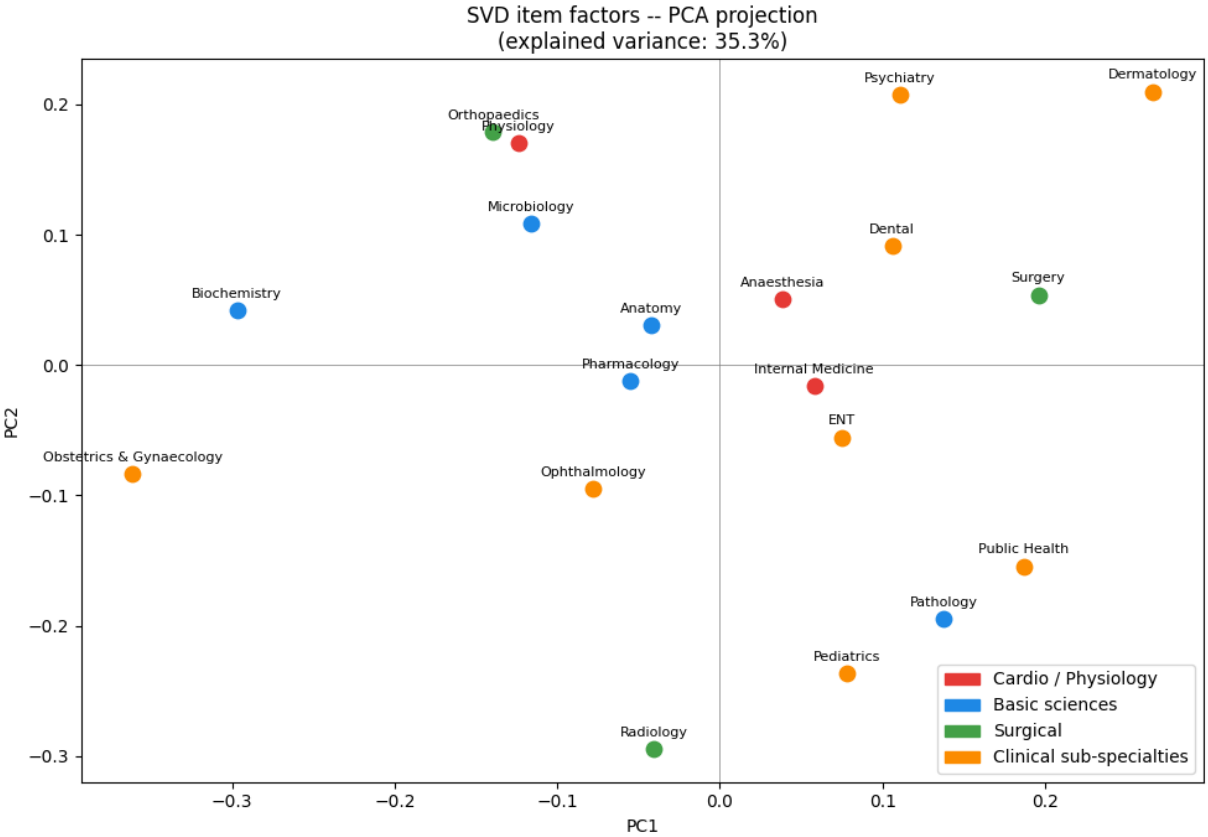

Figure 19: EMMA ablation grid results

Figure 20: Classification champion config error analysis

```
Holdout weighted F1 : 0.6852
Holdout Cohen's kappa: 0.6639

                          precision    recall  f1-score   support

             Anaesthesia       0.74      0.67      0.70       317
                 Anatomy       0.71      0.63      0.67      1456
            Biochemistry       0.74      0.76      0.75       828
                  Dental       0.71      0.64      0.67       894
             Dermatology       0.37      0.25      0.30       177
                     ENT       0.80      0.74      0.77       492
       Internal Medicine       0.59      0.58      0.59      1789
            Microbiology       0.72      0.73      0.72      1131
Obstetrics & Gynaecology       0.80      0.81      0.81      1001
           Ophthalmology       0.80      0.80      0.80       693
            Orthopaedics       0.53      0.48      0.50       300
               Pathology       0.57      0.57      0.57      1489
              Pediatrics       0.65      0.66      0.66       804
            Pharmacology       0.69      0.80      0.74      1376
              Physiology       0.67      0.69      0.68       883
              Psychiatry       0.75      0.79      0.77       444
           Public Health       0.76      0.83      0.79      1778
               Radiology       0.66      0.61      0.63       440
                 Surgery       0.62      0.64      0.63      1686

                accuracy                           0.69     17978
               macro avg       0.68      0.67      0.67     17978
            weighted avg       0.69      0.69      0.69     17978
```

Figure 21: Champion Configuration Confusion Matrix

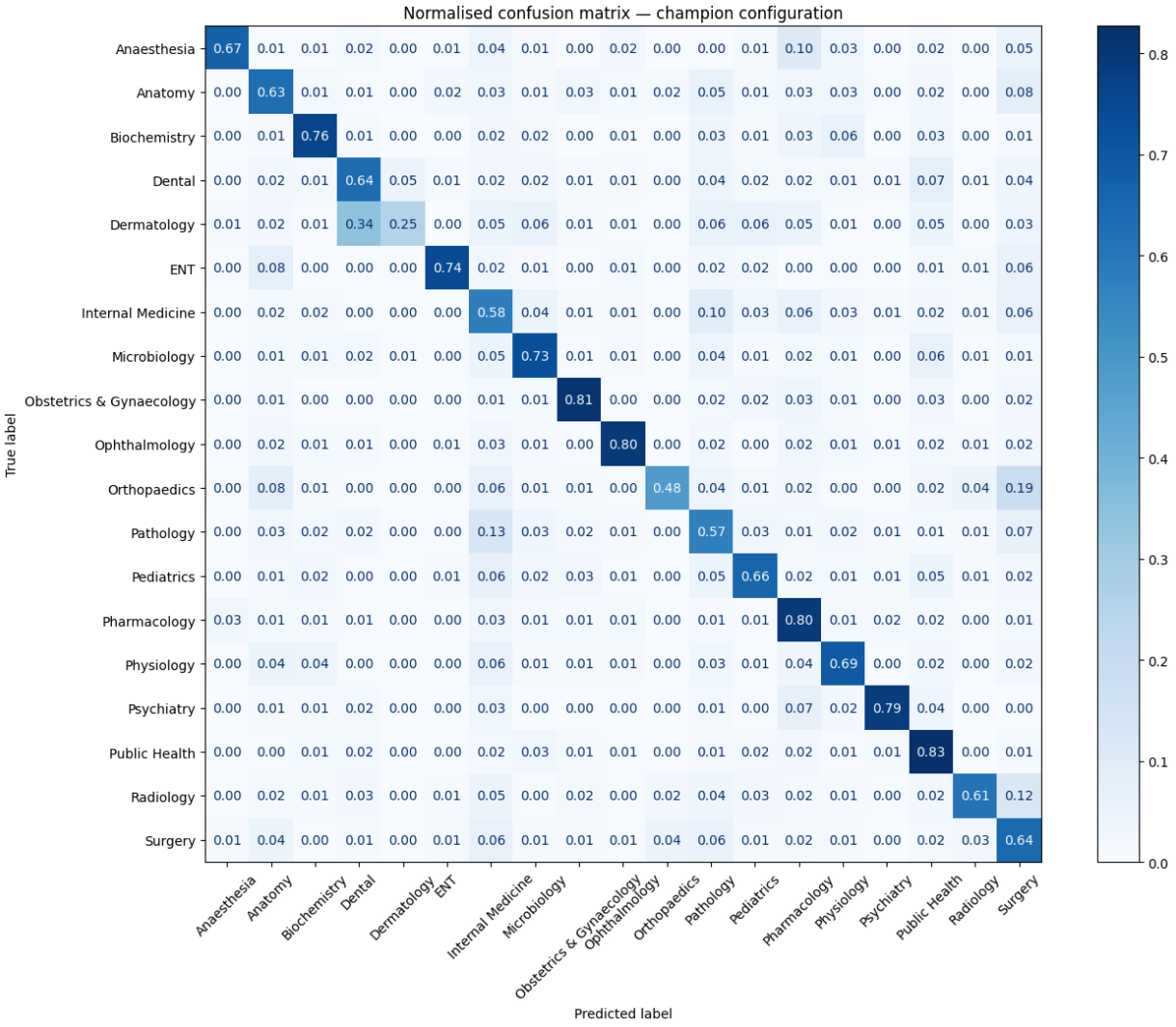

Figure 22: RAG Accuracy by Specialty

	specialty	n	RAG acc%	Base acc%	Delta
0	Internal Medicine	14	28.6	28.6	0.0
1	Pathology	7	42.9	57.1	-14.2
2	Surgery	5	20.0	60.0	-40.0
3	Microbiology	4	50.0	25.0	25.0
4	Psychiatry	4	25.0	25.0	0.0
5	Public Health	3	33.3	66.7	-33.4
6	Pharmacology	3	66.7	66.7	0.0
7	Obstetrics & Gynaecology	3	66.7	33.3	33.4
8	Anatomy	2	0.0	0.0	0.0
9	Dental	2	0.0	50.0	-50.0
10	Pediatrics	2	100.0	50.0	50.0
11	Orthopaedics	1	100.0	100.0	0.0

Tables

Table 1: Full Ablation Grid Results

#	Embedding Model	LLM	RAG	Accuracy	Delta
1	Qwen3-Embedding-0.6B	Qwen3-4B	No	42%	-
2	Qwen3-Embedding-0.6B	Qwen3-4B	Yes	38%	-4pp
3	Qwen3-Embedding-0.6B	Qwen3-4B-Thinking-2507	No	31%	-
4	Qwen3-Embedding-0.6B	Qwen3-4B-Thinking-2507	Yes	32%	+1pp
5	Octen-Embedding-0.6B	Qwen3-4B-Thinking-2507	No	33%	-
6	Octen-Embedding-0.6B	Qwen3-4B-Thinking-2507	Yes	44%	+11pp